QUANT

Time, Speed & Distance

Assignment 02

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Time, Speed & Distance Online Assignment -02

1. A car after travelling 30 km from a point A developed some problem in the engine and the speed became $^4/_5{}^{th}$ of its original speed. As a result, the car reached point B, 45 minutes late. If the engine had developed the same problem after travelling 48 km, then it would have reached B only 36 minutes late. The original speed of the car (in km per hour) and the distance between the points A and B (in km) are

1. 25, 130 2. 20, 150 3. 30, 120 4. None of these

2. Two cyclists do the same journey by travelling at 9 and 10 km/hr respectively. Find the distance travelled when one takes 32 minutes longer than the other.

1. 44 km 2. 48 km 3. 50 km 4. 46 km

3. Joseph walked 1 km/hr slower than usual and he could return home in $^9/_8{}^{th}$ of his usual time. His normal walking rate is

1. 8 km/hr 2. 9 km/hr 3. 10 km/hr 4. 11 km/hr

4. If a man walks at 4 km/hr, he misses the bus by 10 minutes. If he walks at 5 km/hr, he reaches 5 minutes before the arrival of the bus. How far is the bus stand?

1. 10 km 2. 12 km 3. 15 km 4. 5 km

5. Two trains are traveling in opposite directions at 90 kmph and 18 kmph. If the length of the faster train is 600 metre, find the time taken by the faster train to cross a man standing in the slower train.

1. 20 sec. 2. 25 sec. 3. 30 sec. 4. Data inadequate

6. A car travels 192 km at an average speed of x km/hr. A second car travels the same distance at an average speed of $(x + 4)$ km/hr. If the second car arrives 4 hours earlier than the first, the speed of the former is

1. 15 km/hr 2. 12 km/hr 3. 24 km/hr 4. 16 km/hr

7. A car takes 5 hours to travel a certain distance. If it had increased its speed by 8 km/hr, then it would have taken 3 hours. The distance is

1. 200 km 2. 120 km 3. 60 km 4. 40 km

8. A train approaches a tunnel AB. Inside the tunnel a cat is located at a point $^3/_8{}^{th}$ of the distance AB, measured from the entrance A. When the train whistles, the cat runs. If the cat moves to the entrance of the tunnel A, the train catches the cat exactly at the entrance. If the cat moves towards the exit B, the train catches the cat exactly at the exit. The speed of the train is greater than the cat by what order?

1. 3 : 01 2. 4 : 01 3. 5 : 01 4. None of these

9. A man rows 40 km upstream in 8 hours and a distance of 36 km downstream in 6 hours. Then the speed of the man in still water is

1. 0.5 km/hr 2. 5.5 km/hr 3. 6 km/hr 4. 5 km/hr

10. A boat goes 12 km upstream in 48 minutes. The speed of stream is 2 km/hr. The speed of boat in still water is

1. 13 km/hr
2. 2.25 km/hr
3. 17 km/hr
4. 15 km/hr

Answer key & Explanations

Q No.	Key	Explanation
1.	3	The difference of 9 minutes is due to the middle 16 kms, which have been traveled at original speed and at the new speed. Let the original speed of the car be C. Now the equation will be $\frac{18}{4C/5} - \frac{18}{C} = \frac{9}{60}$. Solving you get C = 30 kmph. Now let the remaining distance after traveling 48 km be D. The equation would be $\frac{D}{30 \times \frac{4}{5}} - \frac{D}{30} = \frac{36}{60}$. Solving D = 72 km. Now the total distance will be 72 + 48 = 120 km.
2.	2	Let the distance traveled = x. $\therefore x / 9 - x / 10 = 32 / 60 \Rightarrow x = 48$ km.
3.	2	Let his normal speed = x km/hr. Decreased speed = $(x - 1)$ km/hr. If usual time is t hrs, then on decreasing speed, he takes 9 / 8 of his usual time. But distance traveled on both sides is same. $\therefore \frac{9}{8}(x-1)\,t = tx \Rightarrow x = 9$ km/hr.
4.	4	Let required distance = x km. Difference of time = 5– (–10)=15 min. $\therefore \frac{x}{4} - \frac{x}{5} = \frac{15}{60} \Rightarrow x = 5$ km
5.	1	As the faster train crosses the man in the slower train, time taken in this case = length of the faster train / Relative speed. Thus time = 600 / 30 = 20 seconds.
6.	2	$\frac{192}{x} - \frac{192}{x+4} = 4 \Rightarrow x = 12\text{km/hr}.$
7.	3	Let the distance be x. $\therefore$ speed of train = x / 5. Also distance $x = \left(\frac{x}{5} + 8\right)3 \Rightarrow x = 3x / 5 + 24$ $\Rightarrow 2x / 5 = 24 \Rightarrow x = 60$ km.
8.	1	Let the dist of the train from the tunnel be x and the length of the tunnel be y. Ratio of speeds in case 1 is x : 3/8 y. In the second case it is x + y : 5/8 y. Equating the two we get the ratio x : y as 3 : 2. And the ratio of the speeds = x : 3/8 y = 4 : 1. But the question is asking about greater, now the speed is greater in the ratio of 3 : 1
9.	3	Speed of upstream = 40 / 8 = 5 km / hr. Speed of downstream = 36 / 6 = 6 km / hr. $\therefore$ speed of man in still water = (5 + 6) / 2 = 5.5 km / hr.
10.	3	12 km upstream in 48 min. $\therefore$ it will cover 15 km in 1 hr. Speed of stream = 2 km / hr. $\therefore$ speed of boat in still water = 15 + 2 = 17 km / hr.